



Forecasting Conflict Fatalities with a Grievance-Based Model

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Abstract

This paper evaluates whether a grievance-based model—defined as a framework capturing deprivation, injustice, exclusion, discrimination, and perceived unfairness [1]—can improve forecasts of conflict fatalities using Random Forest (RF) and Gradient Boosting Machine (GBM). Conflict fatalities are the outcome of events where “an incident where armed force was used by an organised actor against another organized actor, or against civilians, resulting in at least 1 direct death” [2]. We operationalize grievance through fifteen indicators drawn from Freedom House [3] and Varieties of Democracy (V-Dem) [4]. Neither model outperforms the benchmark models. Code https://github.com/Brinkley97/pos-6933_conflict_forecasting

1. Introduction

Words of an individual or group can be empty; thus, we must examine their actions. Words of the United States Declaration of Independence on July 4, 1776, famously states, “We hold these truths to be self-evident, that all men are created equal, that they are endowed by their Creator with certain unalienable Rights, that among these are Life, Liberty and the pursuit of Happiness.” Yet, the actions of the US, at this time, were to enslave Black people. This profound contradiction led to numerous conflicts, most notably the Nat Turner Rebellion in August 1831 and the American Civil War, which ended in May 1865 and ultimately resulted in the emancipation of enslaved people. Following these foundational grievances and catastrophic conflicts, this memo is a generalization of using grievance-based predictors (also called indicators) to forecast conflict on a global scale.

To this notion, we use fifteen indicators with two from the Freedom House [3] and thirteen from Varieties of Democracy (V-Dem) [4] datasets to forecast conflict. We use two machine learning: Random Forest (RF) and Gradient Boosting Machine (GBM), then compare their out-of-sample predictive performance against the two benchmark models provided for this course. Neither one of our models outperforms the benchmark models.

2. Theory and Operationalization

We selected Freedom House and Varieties of Democracy (V-Dem) and didn’t include the World-wide Governance Indicators (WGI), Polity Dataset, Quality of Government (QoG), World Development Indicators (WDI), or Georeferenced Event Dataset (GED) because those datasets do not adequately capture deprivation, injustice, exclusion, discrimination, and perceived unfairness.

2.1 Freedom House (FIW)

The main content for this dataset is political rights/civil liberties covering the years of 1990 to 2024 [5] for a selection of entities (countries and territories). More specifically, this dataset maps political rights and civil liberties to numerical ratings and descriptive texts. For numerical ratings, the score ranges between 0 to 4 points. There exists 10 political rights indicators and 15 civil liberties indicators. So, 10 indicators can have a low of 0 (0 points $\times$ 10 indicators) and high of 40 (4 points $\times$ 10 indicators) and 15 indicators can have a low of 0 (0 points $\times$ 15 indicators) and high of 60 (4 points $\times$ 15 indicators). The lower the score corresponds to less degree of freedom, while the higher, the greatest degree of freedom. They also are used to derive the status—Free, Partly Free, and Not Free (see Figure 1).

A country or territory's *Freedom in the World* status depends on its aggregate Political Rights score, on a scale of 0–40, and its aggregate Civil Liberties score, on a scale of 0–60. The total Political Rights and Civil Liberties scores are equally weighted in this calculation, leading to the following possible ranges.

Status		Political Rights score						
		0-5*	6-11	12-17	18-23	24-29	30-35	36-40
Civil Liberties score	53-60	PF	PF	PF	F	F	F	F
	44-52	PF	PF	PF	PF	F	F	F
	35-43	PF	PF	PF	PF	PF	F	F
	26-34	NF	PF	PF	PF	PF	PF	F
	17-25	NF	NF	PF	PF	PF	PF	PF
	8-16	NF	NF	NF	PF	PF	PF	PF
	0-7	NF	NF	NF	NF	PF	PF	PF

F = Free, PF = Partly Free, and NF = Not Free

* It is possible for a country or territory's total political rights score to be less than zero (between -1 and -4) if it receives mostly or all zeros for each of the 10 political rights questions and it receives a sufficiently large negative score for the political rights discretionary question.

Figure 1: Image of scores table [3]

Table 1: Mapping Freedom House Indicators to Grievance-Related Questions

Variable	Related Question from Freedom House Methodology [3]	Connection to Forecasting Conflict Fatalities
fh_pr (Political Rights)	B2: Is there a realistic opportunity for the opposition to increase its support or gain power through elections?	A low score here signifies blocked political channels. When groups cannot achieve power peacefully, they may resort to violence, increasing the risk of conflict fatalities.
	B4: Do various segments of the population (including ethnic, racial, religious, gender, LGBT+, and other relevant groups) have full political rights and electoral opportunities?	A low score indicates systemic exclusion of specific groups. This creates a clear grievance, motivating collective action against the state and heightening the likelihood of violence.
	C2: Are safeguards against official corruption strong and effective?	A low score suggests that state resources are being captured by elites, leading to public deprivation and a perception of profound injustice—core components of the Grievance Model [1].
	ADS: Is the government or occupying power deliberately changing the ethnic composition of a country or territory so as to destroy a culture or tip the political balance in favor of another group?	This represents an extreme form of group-based grievance and existential threat, acting as a powerful motivator for armed resistance and intercommunal violence.
fh_cl (Civil Liberties)	G4: Do individuals enjoy equality of opportunity and freedom from economic exploitation?	A low score directly measures economic grievance. A lack of opportunity and presence of exploitation creates a population with little to lose from conflict.

Note: Question codes reference the subcategories from the Freedom House methodology [3]: B = Political Pluralism and Participation, C = Functioning of Government, G = Personal Autonomy and Individual Rights, ADS = Additional Discretionary Question.

2.2 Varieties of Democracy (V-Dem)

The main content of this dataset is a democracy measure covering the years 1990 to 2024 [5] for a selection of countries. More specifically, this dataset moves past elections only to adhere to multiple dimensions while distinguishing between five high-level principles of democracy: electoral, liberal, participatory, deliberative, and egalitarian [6].

Table 2: Mapping V-Dem Equality & Exclusion Indicators to Grievance Theory

Variable	Concept Captured	Connection to Forecasting Conflict Fatalities
vdem_v2x_egaldem (2.1.5)	An interval score from 0 (low) to 1 (high) where rights, resources, and access to power are distributed equally across all social groups.	A lower score signifies systemic inequality, which may increase the chances of conflict by marginalized groups.
vdem_v2x_egal (2.2.18)	This index is an average of the equal protection, equal access, and equal resource distribution indices.	A low score indicates broad-based inequality, a core driver of the Grievance Model [1].
vdem_v2xeg_eqprotec (2.2.19)	An interval score from 0 (low) to 1 (high) measuring if the state grants and protects rights evenly across social groups.	When a social group does not feel protected by the state, its members may see violence as their only recourse.
vdem_v2xeg_eqaccess (2.2.20)	An interval score from 0 (low) to 1 (high) measuring if all groups have equal de facto capabilities to participate in and influence politics.	A low score indicates political exclusion. If groups cannot influence policy through formal channels, they may turn to conflict.
vdem_v2xeg_eqdr (2.2.21)	An interval score from 0 (low) to 1 (high) measuring the equality of resource distribution.	A low score signifies economic deprivation for certain groups, a powerful motivator for violent mobilization.

Note: Codes reference the subcategories from the V-Dem codebook [4]. Equality & Exclusion Indicators: 2.1.5 = Egalitarian democracy index; 2.2.18 = Egalitarian component index; 2.2.19 = Equal protection index; 2.2.20 = Equal access index; 2.2.21 = Equal distribution of resources index.



Table 3: Mapping V-Dem Voice & Association Indicators to Grievance Theory

Variable	Concept Captured	Connection to Forecasting Conflict Fatalities
vdem_v2x_freexp_altinf (2.2.3)	To what extent does government respect press and media freedom, the freedom of ordinary people to discuss political matters at home and in the public sphere, as well as the freedom of academic and cultural expression?	A low score indicates that non-violent channels for expressing grievances are blocked, making violence a more likely option.
vdem_v2x_frassoc_thick (2.2.4)	To what extent are parties, including opposition parties, allowed to form and to participate in elections, and to what extent are civil society organizations able to form and to operate freely?	A low score signifies that citizens cannot organize peacefully to address their grievances, potentially forcing them toward covert and violent mobilization.
vdem_v2x_suffr (2.2.5)	What share of adult citizens as defined by statute has the legal right to vote in national elections?	A low score is a direct measure of political exclusion, a primary component of grievance theory [1].
vdem_v2x_cspart (2.2.13)	Are major CSOs routinely consulted by policymakers; how large is the involvement of people in CSOs; are women prevented from participating; and is legislative candidate nomination within party organization highly decentralized or made through party primaries?	A low score indicates a weak civil society, meaning fewer non-state avenues exist for mediating disputes and channeling grievances peacefully.

Note: Codes reference the subcategories from the V-Dem Codebook [4]: 2.2.3 = Freedom of Expression and Alternative Sources of Information index; 2.2.4 = Freedom of association thick index; 2.2.5 = Share of population with suffrage; 2.2.13 = Civil society participation index.



Table 4: Mapping V-Dem Electoral Unfairness Indicators to Grievance Theory

Variable	Concept Captured	Connection to Forecasting Conflict Fatalities
vdem_v2elrfair (3.1.3.15)	Taking all aspects of the pre-election period, election day, and the post-election process into account, would you consider this national election to be free and fair?	A low score delegitimizes the political system. If elections are perceived as rigged, groups may conclude that violence is the only way to achieve political change.
vdem_v2elvotbuy (3.1.3.7)	In this national election, was there evidence of vote and/or turnout buying?	Vote buying signals systemic corruption and undermines democratic legitimacy, fueling the perception of an unfair system which is central to the Grievance Model [1].
vdem_v2elirreg (3.1.3.8)	In this national election, was there evidence of other intentional irregularities by incumbent and/or opposition parties, and/or vote fraud?	This is a direct measure of institutional injustice. High levels of irregularities erode trust and can trigger post-election conflict.
vdem_v2elintim (3.1.3.9)	In this national election, were opposition candidates/parties/campaign workers subjected to repression, intimidation, violence, or harassment by the government, the ruling party, or their agents?	This represents a form of repression that creates grievances and can provoke violent backlash from opposition groups.

Note: Codes reference the subcategories from the V-Dem codebook [4]. Electoral Unfairness Indicators: 3.1.3.15 = Election free and fair; 3.1.3.7 = Election vote buying; 3.1.3.8 = Election other voting irregularities; 3.1.3.9 = Election government intimidation.

3. Data and Research Design

The outcome variable is logged conflict fatalities at seven future horizons, measured by the variables `ln_ged_best_sb_s1` through `ln_ged_best_sb_s7` [7]. The analysis follows the standard course train/test split: training origins span month IDs 1–371 (January 1990– December 2020) and testing targets span month IDs 373–419 (January 2021–December 2024) [8].

We estimate two theory-driven models based on the Grievance framework:

- **Grievance Random Forest (RF):** RF is derived from decision trees, where the core weakness of a single decision tree is its tendency to overfit the training data. RF addresses this by training an ensemble of trees [9] to prevent overfitting while preserving the ability to detect



nonlinearities and complex interaction effects among predictors [10].

- **Grievance Gradient Boosting Machine (GBM):** GBM processes sequentially, with each new tree trained on the errors of the previous tree to reduce errors. This error-correcting strategy reduces bias rather than variance, making GBM complementary to RF within a diverse ensemble [11].

Following [12], these models are evaluated against two benchmark specifications:

- a lag-based benchmark random forest
- a rolling-average benchmark random forest.

Performance is assessed using matched prediction rows and standard out-of-sample metrics: mean squared error (MSE), root mean squared error (RMSE), deterministic CRPS, and probabilistic CRPS. We also examine 80% interval coverage and interval sharpness to assess the quality of the uncertainty estimates.

4. Results

In Figure 2, the Grievance GBM consistently outperforms the Grievance Random Forest, suggesting that the boosting algorithm is better able to extract the predictive signal that exists in the grievance indicators [11]. This finding is consistent with the example Democracy Model memo [12]. However, neither theory-driven model outperforms the benchmark models. The lag-based benchmark, which uses recent conflict history as its main predictor, remains the strongest specification across all forecast horizons and evaluation metrics (MSE, RMSE, and CRPS).

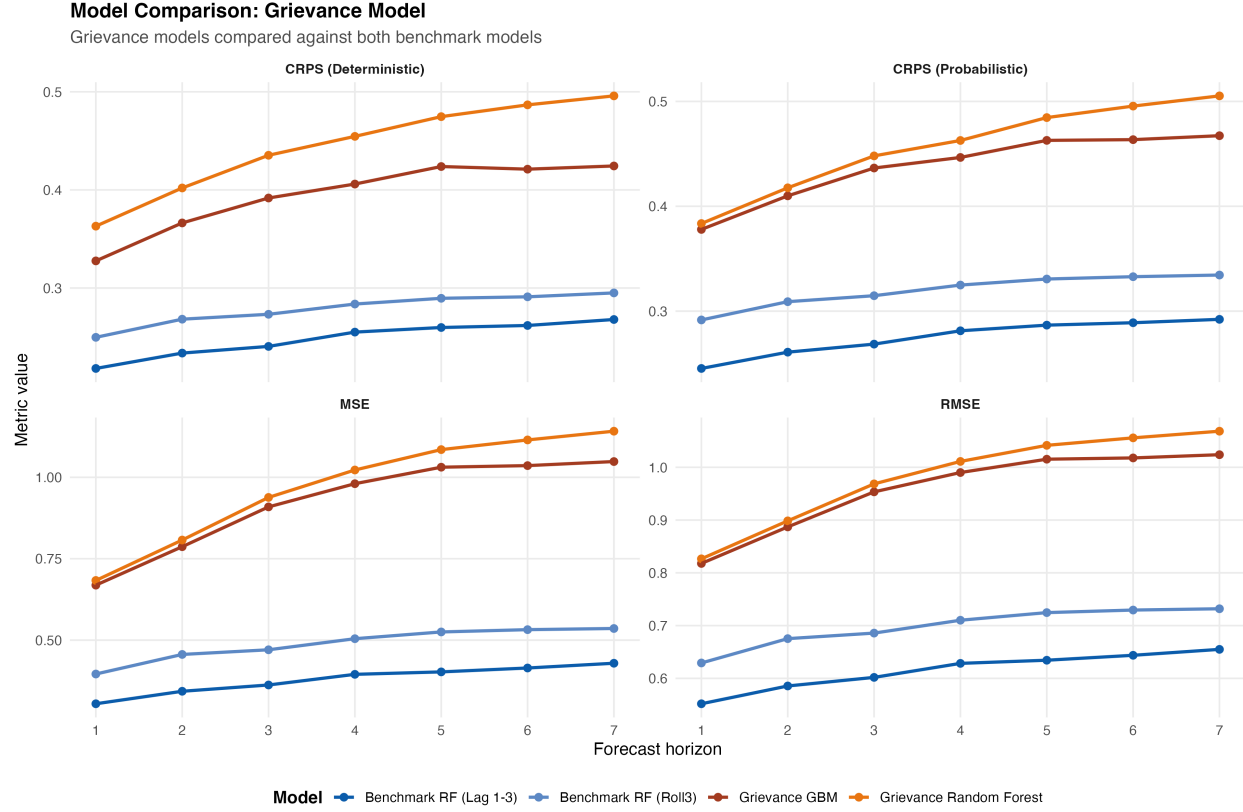


Figure 2: Model performance across all four metrics and seven forecast horizons. Lower values indicate better performance.

5. Variable Importance

When assessing the top predictors, `vdem.v2xeg_eqdr` emerges as the leading predictor for GBM and `vdem.v2x_egaldem` for RF.

`vdem.v2xeg_eqdr` measures the equality of resource distribution as an interval score from 0 (low) to 1 (high) (see Table 2). According to V-Dem [4], the index is formed by taking the point estimates from a Bayesian factor analysis model of the indicators for particularistic or public goods (`v2dlencmps`), means-tested vs. universalistic welfare policies (`v2dlunivl`), educational equality (`v2peedueq`), and health equality (`v2pehealth`). Thus, it is a leading indicator because groups denied equitable access to resources have stronger incentives to resort to violence as a means of redistribution or political change.

`vdem.v2x_egaldem` measures the egalitarian democracy index as an interval score from 0 (low) to 1 (high) (see Table 2). According to V-Dem [4], the index is anchored to the level of electoral democracy (`v2x_polyarchy`) and the egalitarian component index (`v2x_egal`), which averages equal protection, equal access, and equal resource distribution across social groups. Thus, `vdem.v2x_egaldem` is a leading indicator because systemic inequality across protection, access, and resources signals the structural conditions under which marginalized groups are most likely to turn to violence.

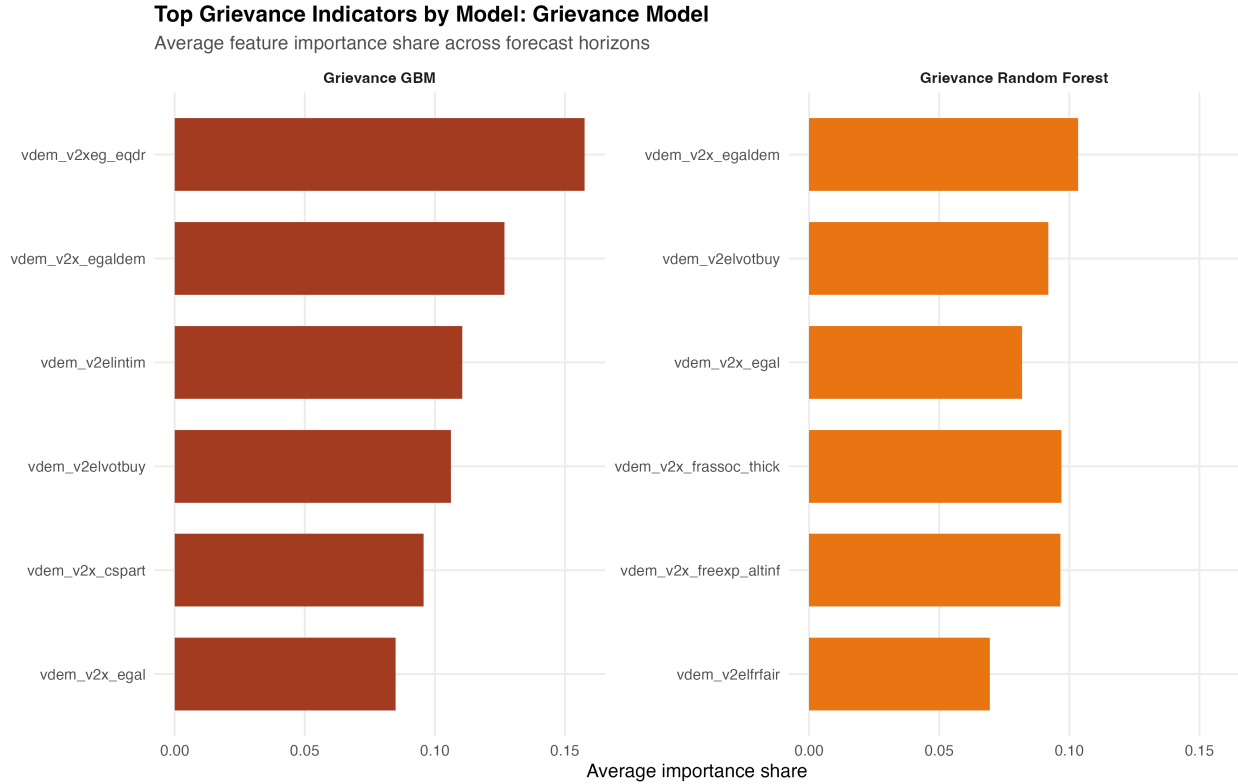


Figure 3: Top Grievance predictors by average importance share across forecast horizons.

6. Conclusion

This memo accentuated the Grievance theory-driven model, fifteen predictors used, two models, and metrics. We motivated the Grievance theory by the foundational argument that deprivation, injustice, exclusion, discrimination, and perceived unfairness are core drivers of political violence [1]—as illustrated historically by the Nat Turner rebellion and the Civil War. We find that the Grievance model performs worse compared to baseline models (Lag 3 and Roll 3), which captures the momentum of recent violence. The main two takeaways are GBM and the top predictors, we see that `vdem_v2xeg_eqdr` is top for GBM and `vdem_v2x_egal` is top for RF. Although grievance indicators retain substantive value, their predictive contribution remains weaker than the autoregressive signal embedded in the two benchmark models.

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